

0050 Fundamental Model v0.1 — Stage B Predictive Validation

Research question

Do the already-Frozen Quality, Fundamental State, state_detail and Valuation categories contain information about later adjusted-close returns?

Frozen data contract

Stage A signal SHA-256: 9c13f87abe8ca25efeb869482724eb49f89c1c234df240d72472641e6bd81241. Entry is exactly Stage A first_trade_date. Horizons are 60, 120, 252 and 504 common TWSE trading sessions.

Cohort boundary

Only the 38 non-financial issuers in the fixed September 2026 cohort enter the analysis. Twelve financial issuers are excluded. The claim is current fixed cohort predictive association only.

Eligible returns

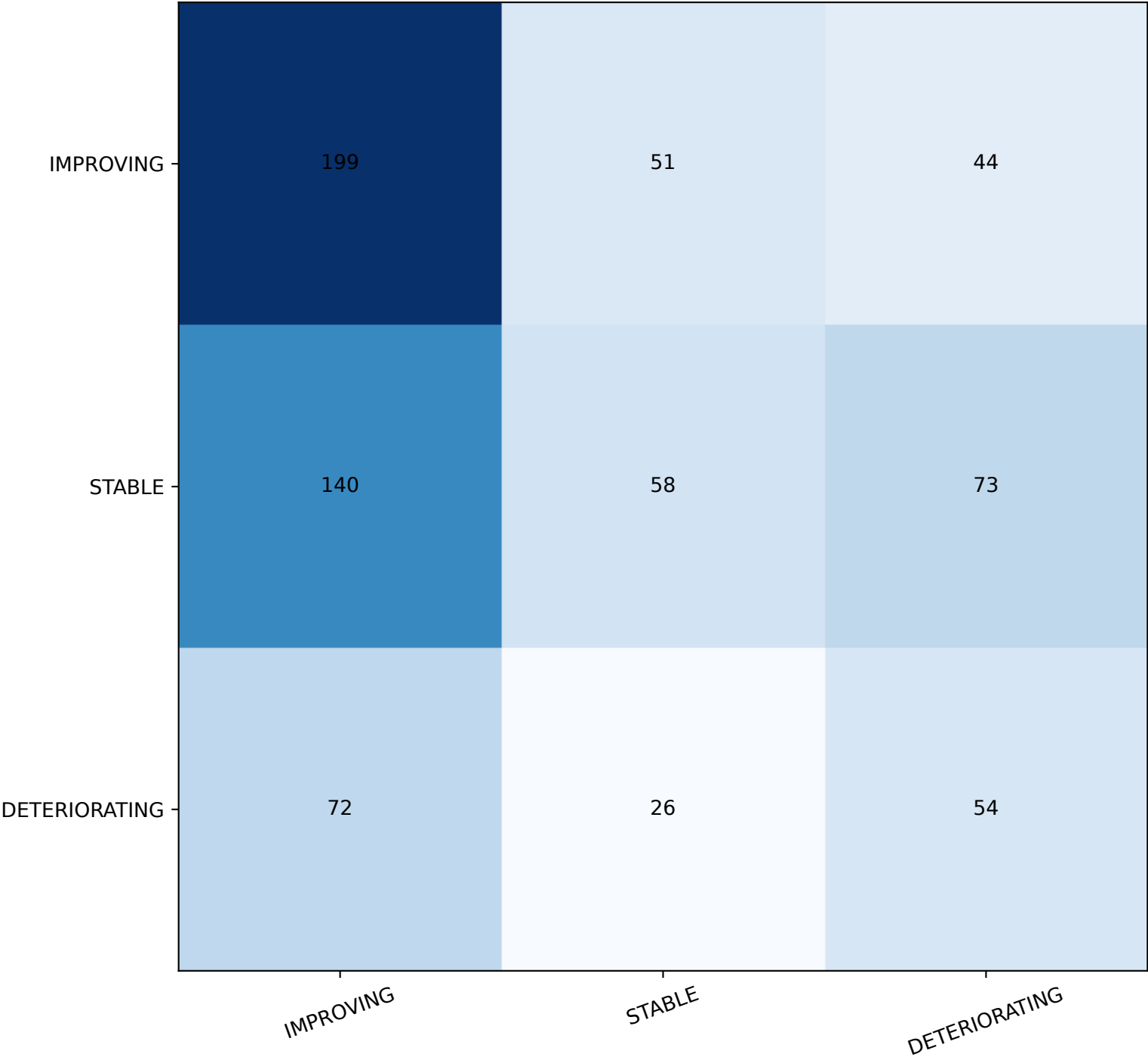
60d: 1502, 120d: 1457, 252d: 1390, 504d: 1219

Conclusion

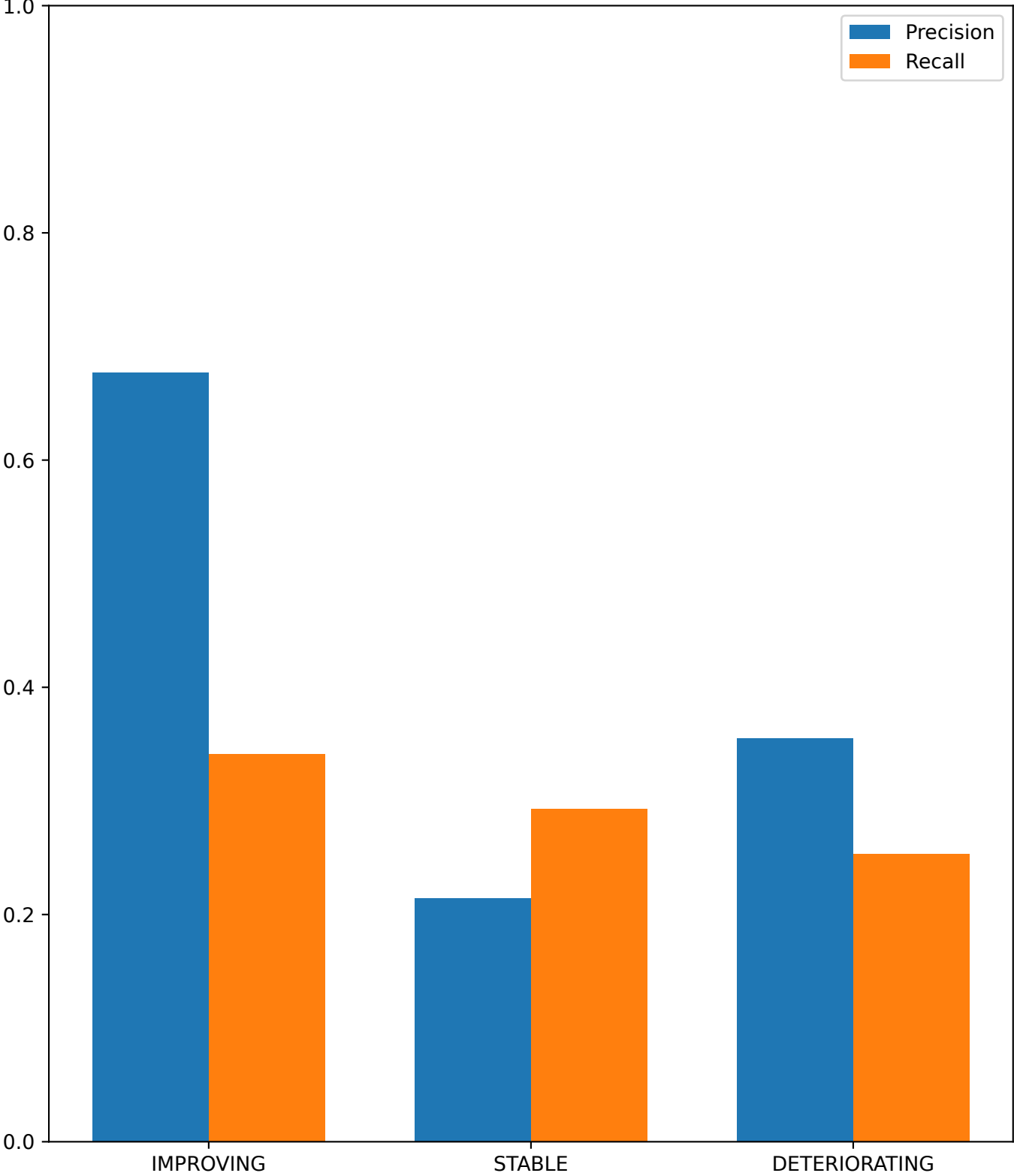
Evidence grade: STRONG. Label: EXPLORATORY_PREDICTIVE_ASSOCIATION. A correct Stage B can PASS even when evidence is weak or none.

A. Fundamental Identification Accuracy

Frozen identification confusion matrix



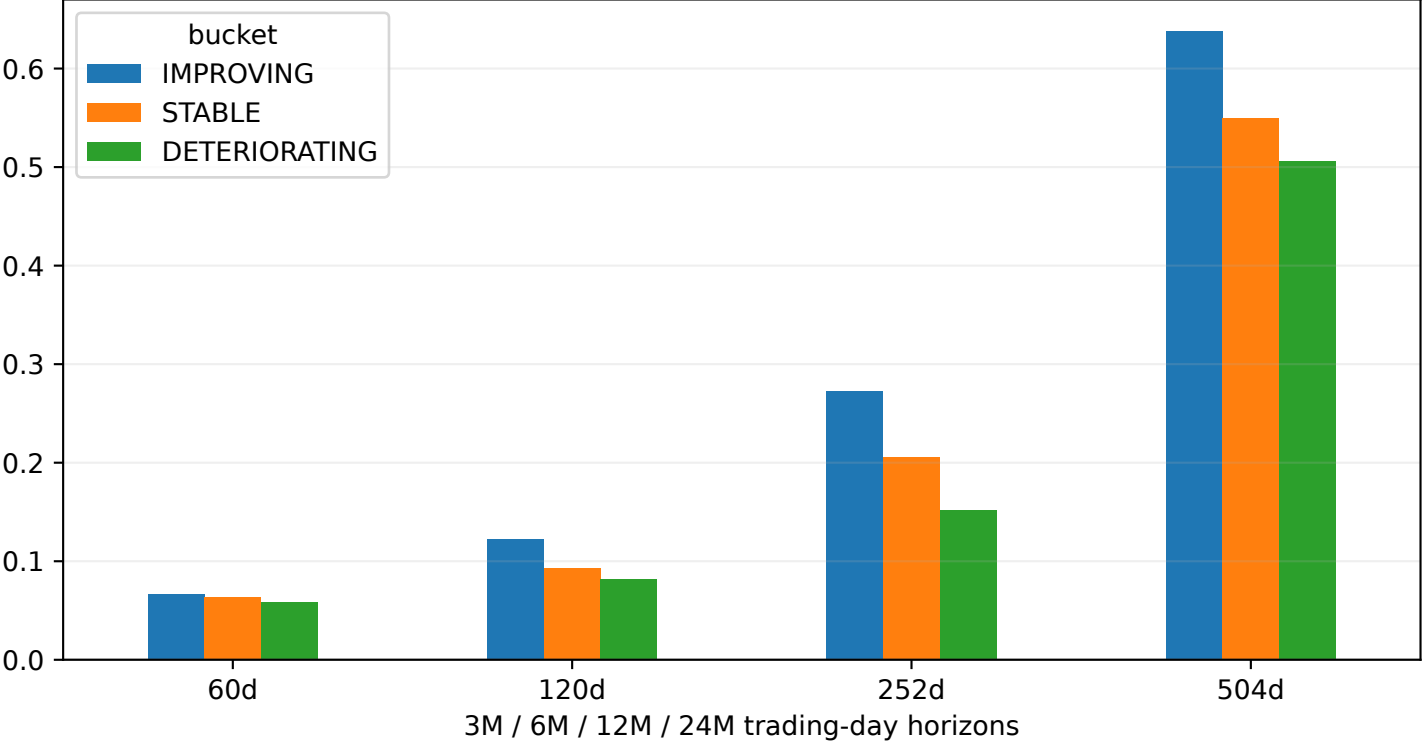
Frozen canonical-state accuracy



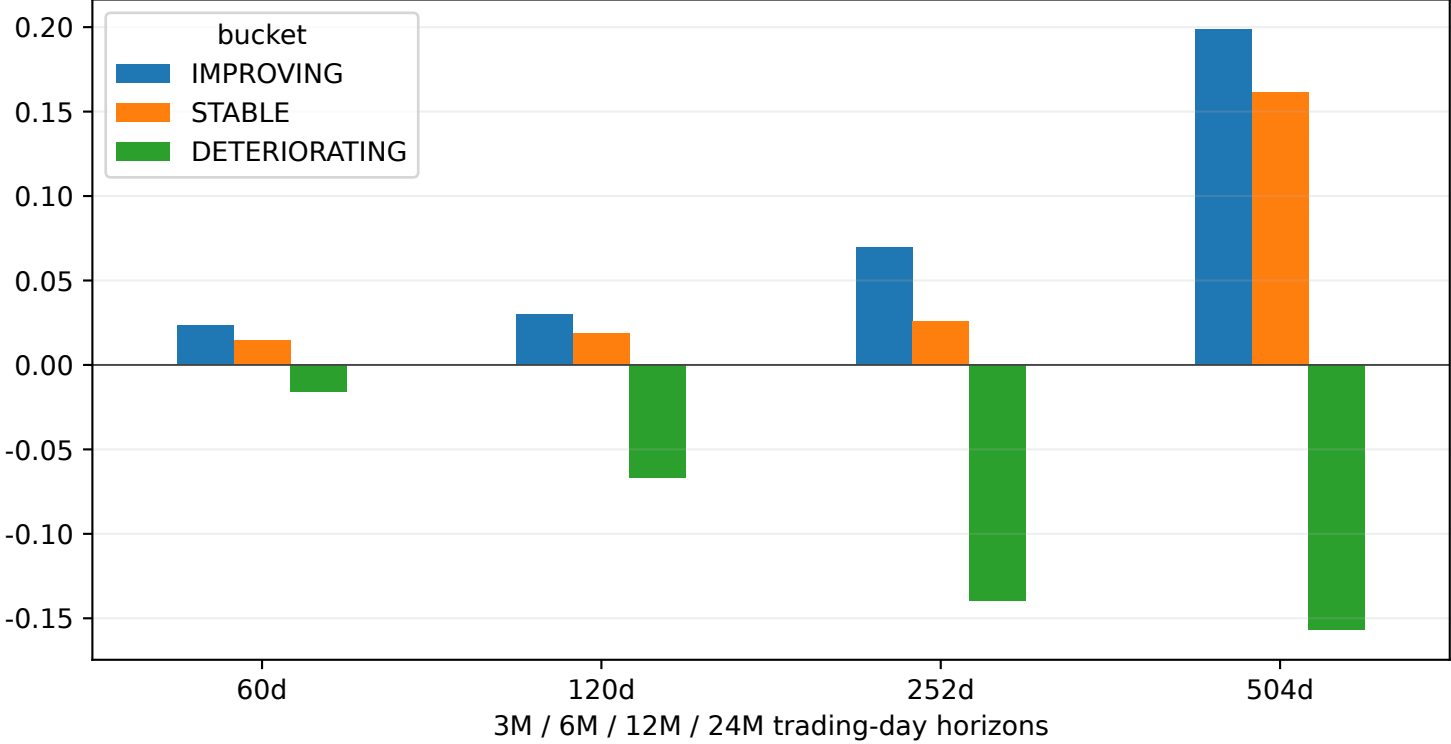
Identification accuracy and future-return predictive power are separate research layers.

B1. Future Return Predictive Power — Canonical State

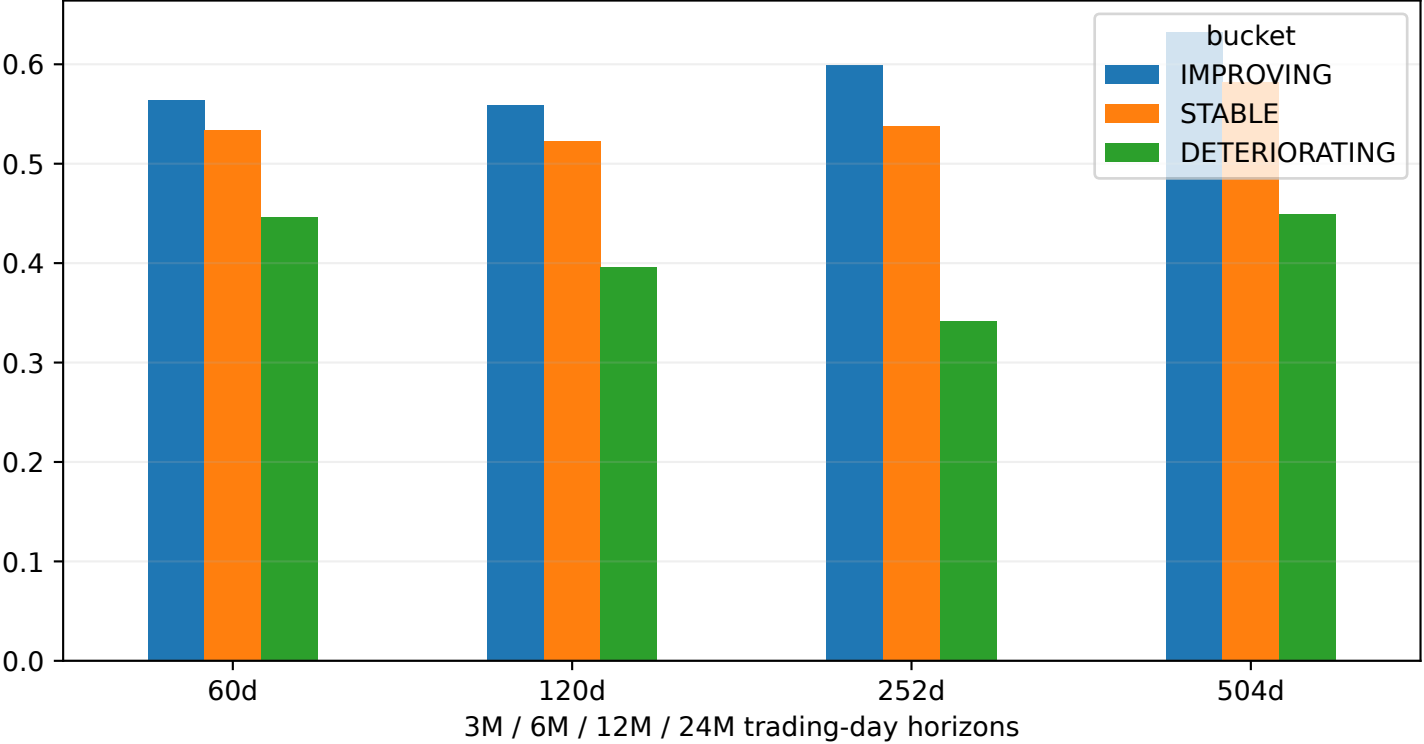
Median stock return



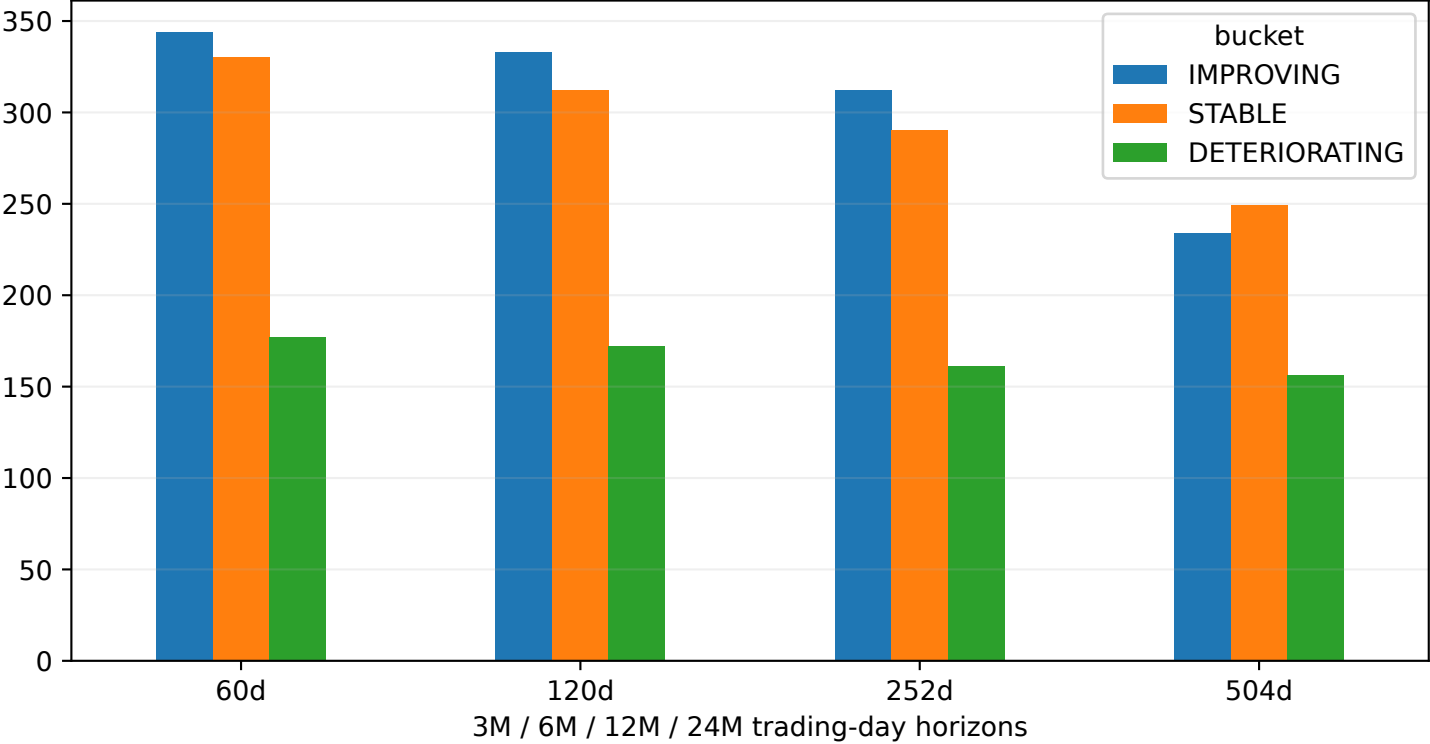
Median excess return vs 0050



Outperform-0050 rate



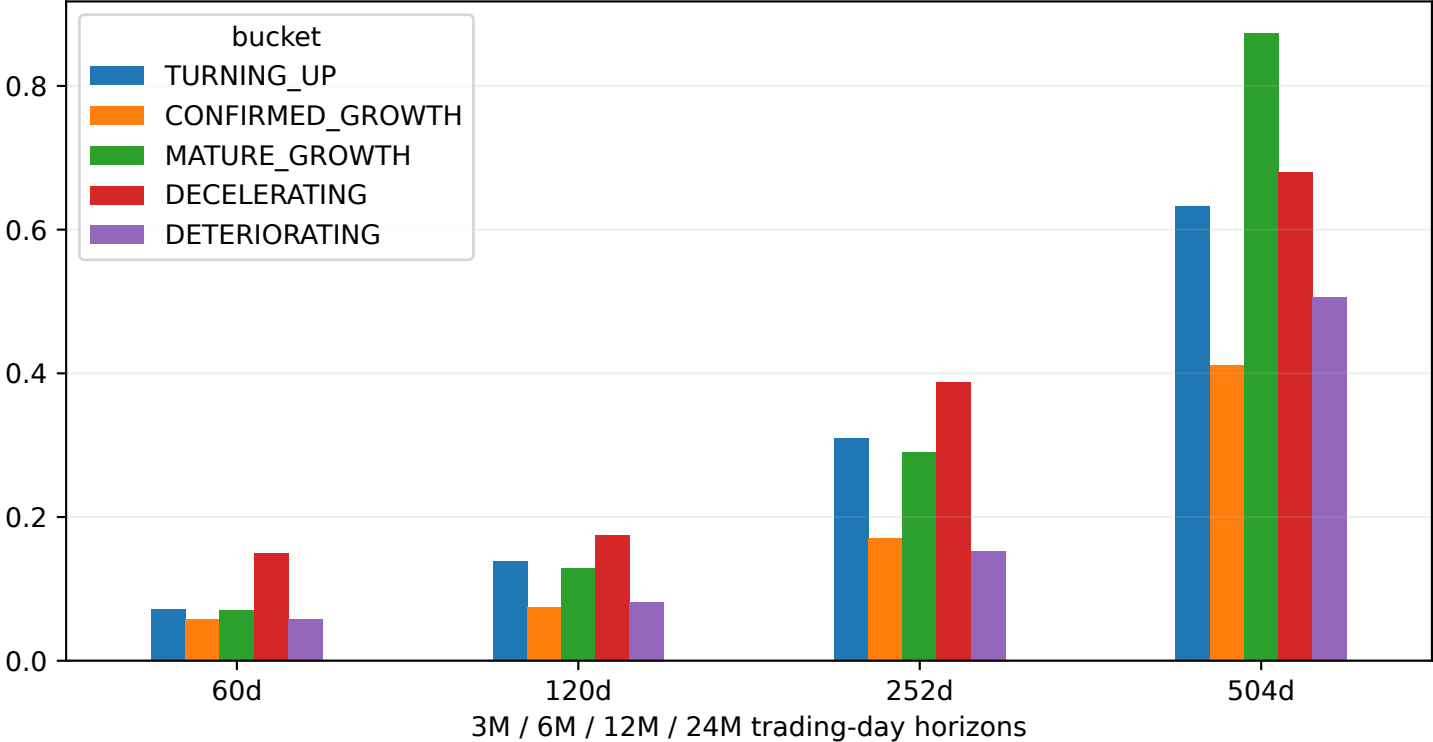
Evaluable observations



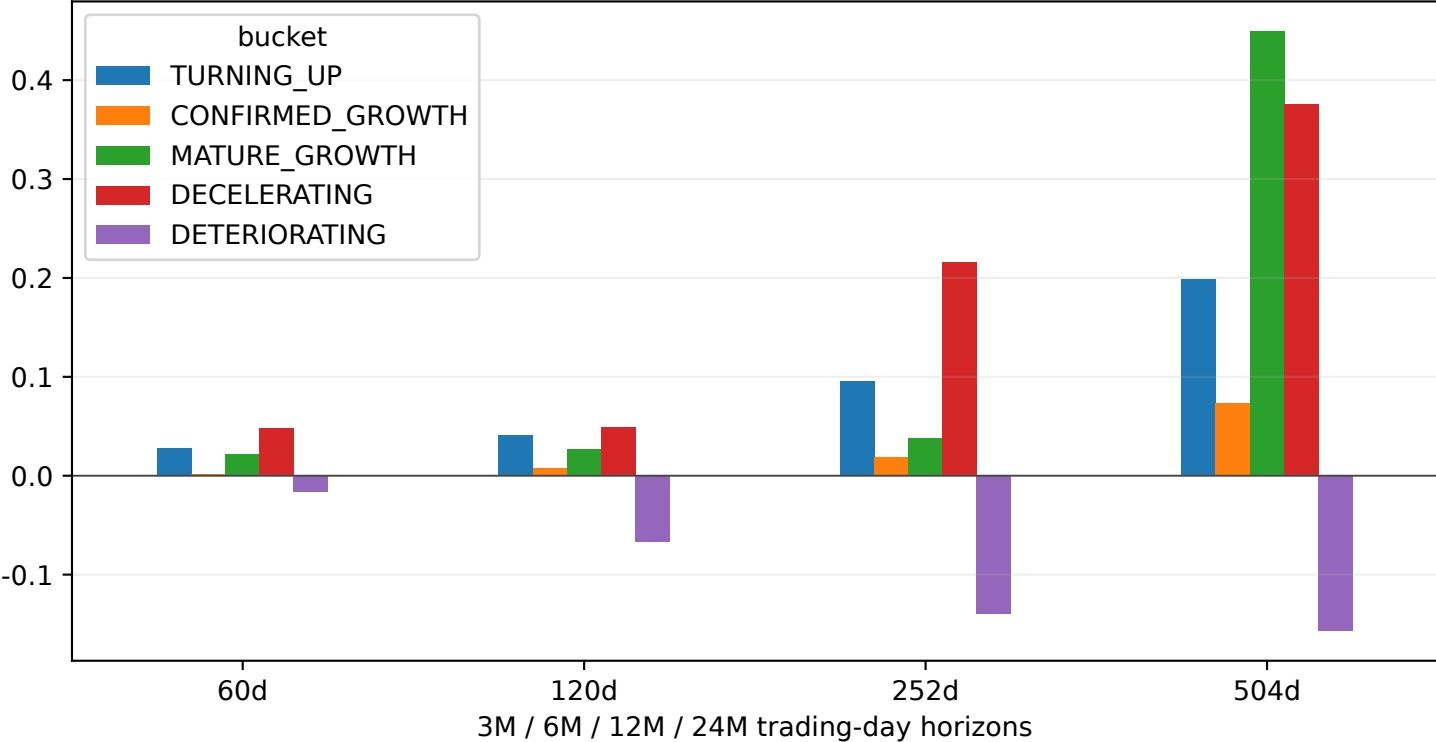
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B2. Future Return Predictive Power — state_detail

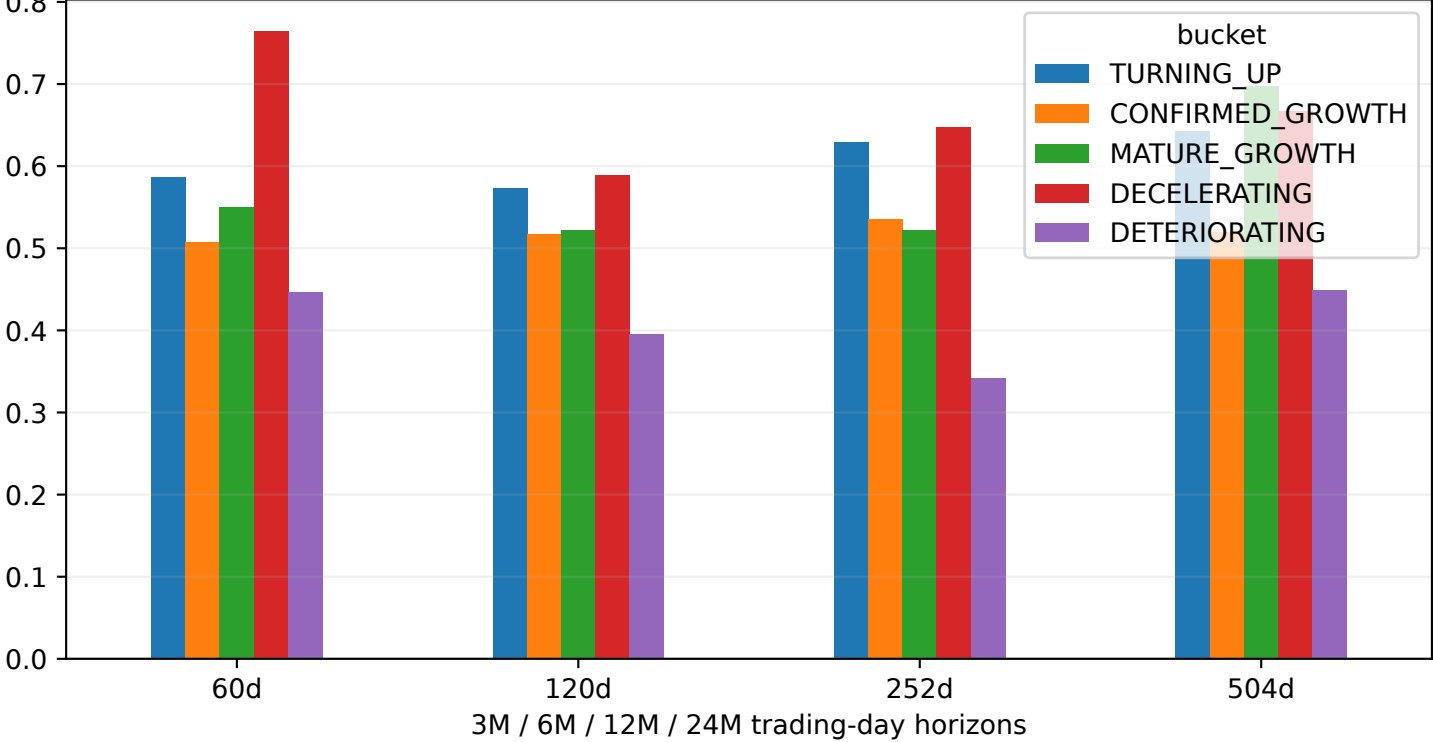
Median stock return



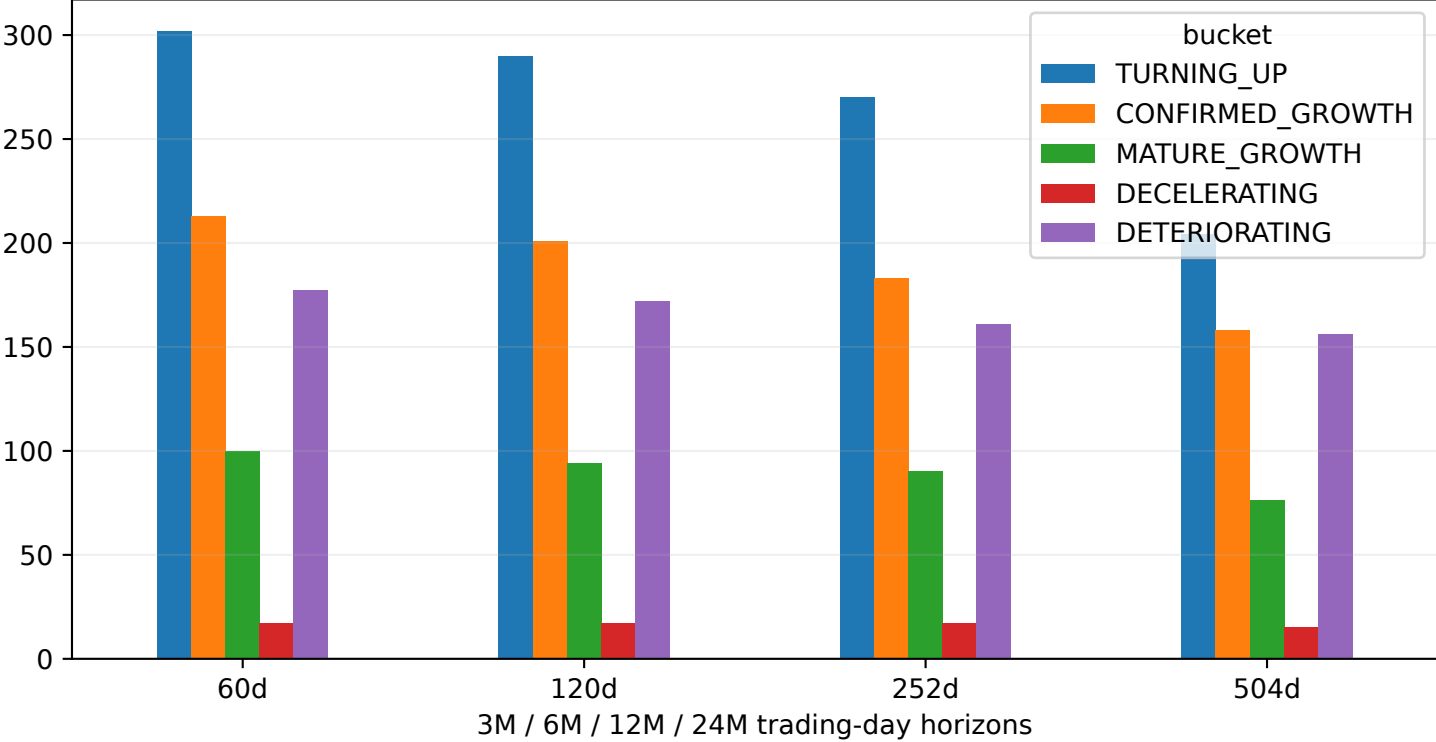
Median excess return vs 0050



Outperform-0050 rate



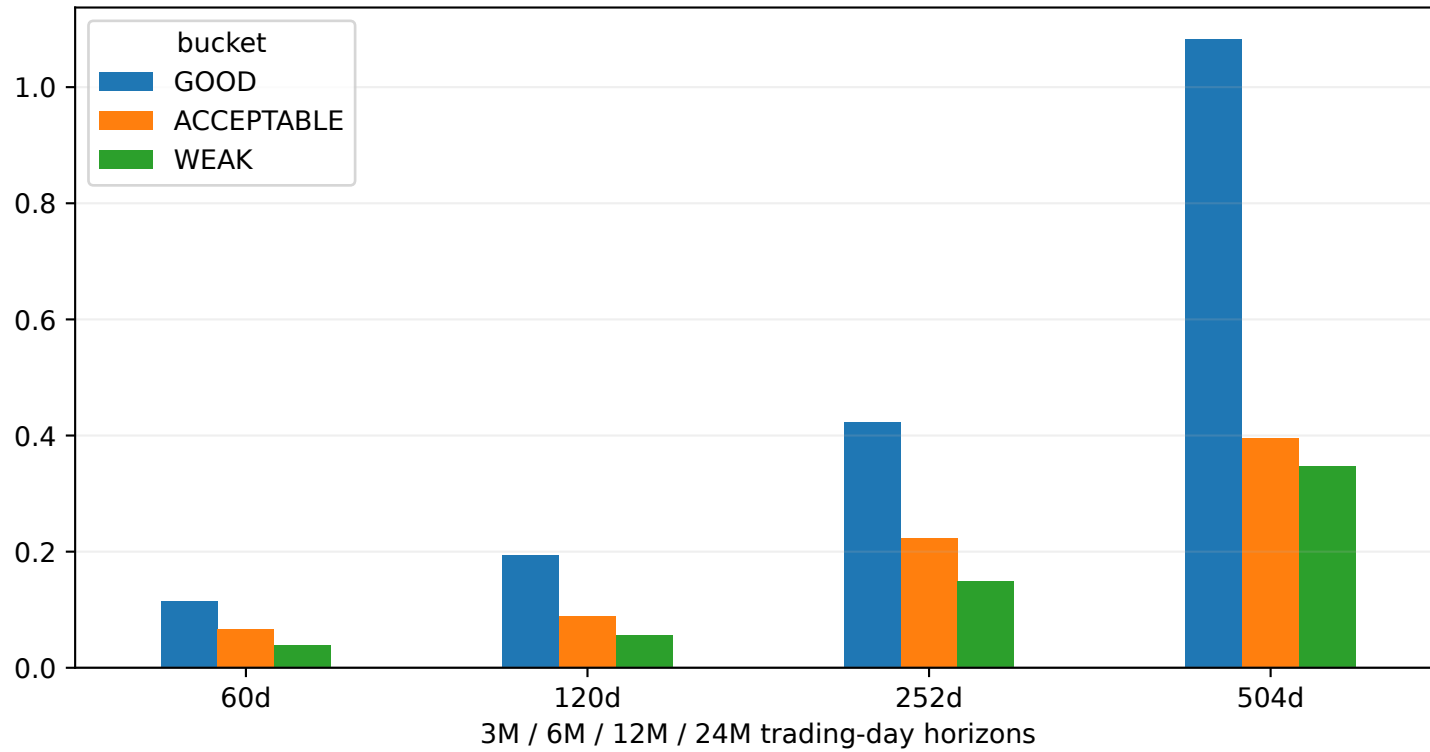
Evaluable observations



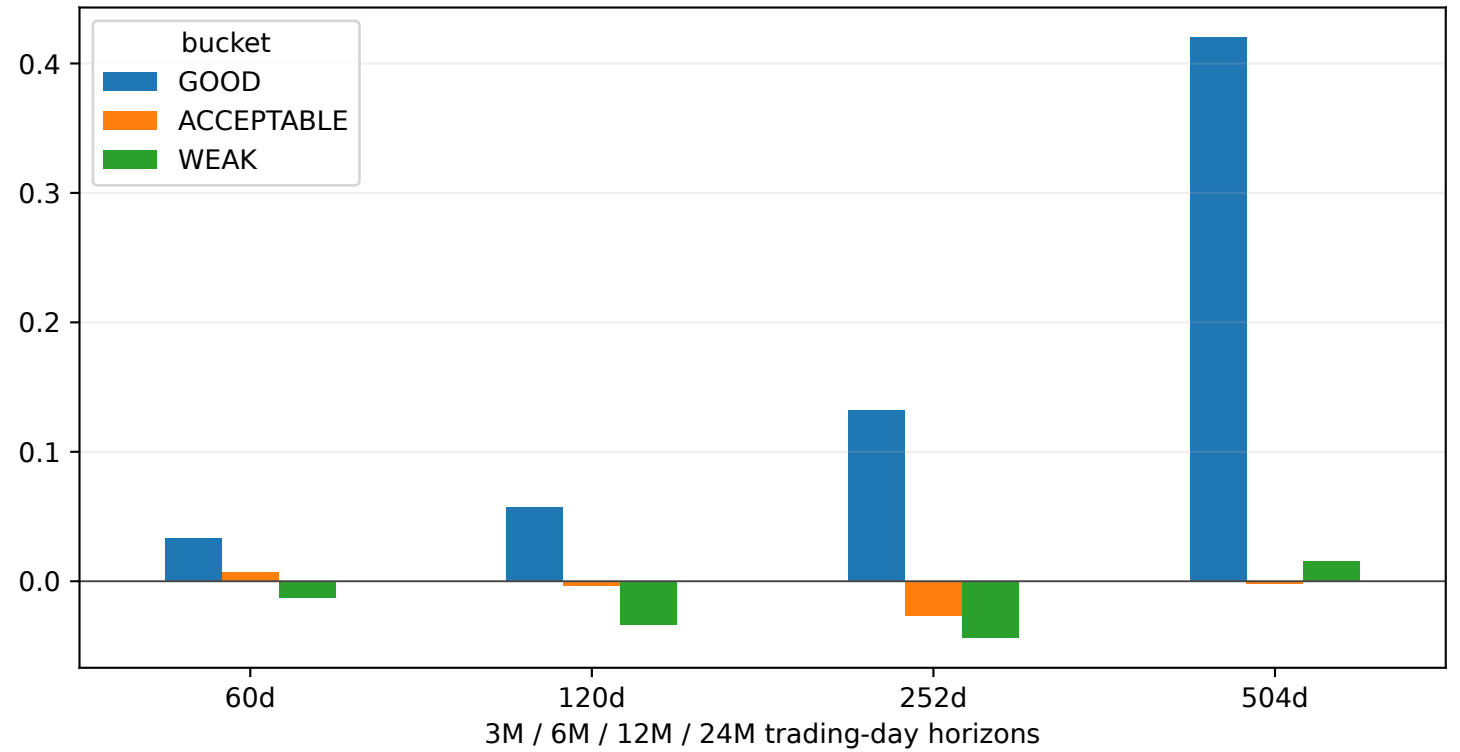
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B3. Quality Predictive Test

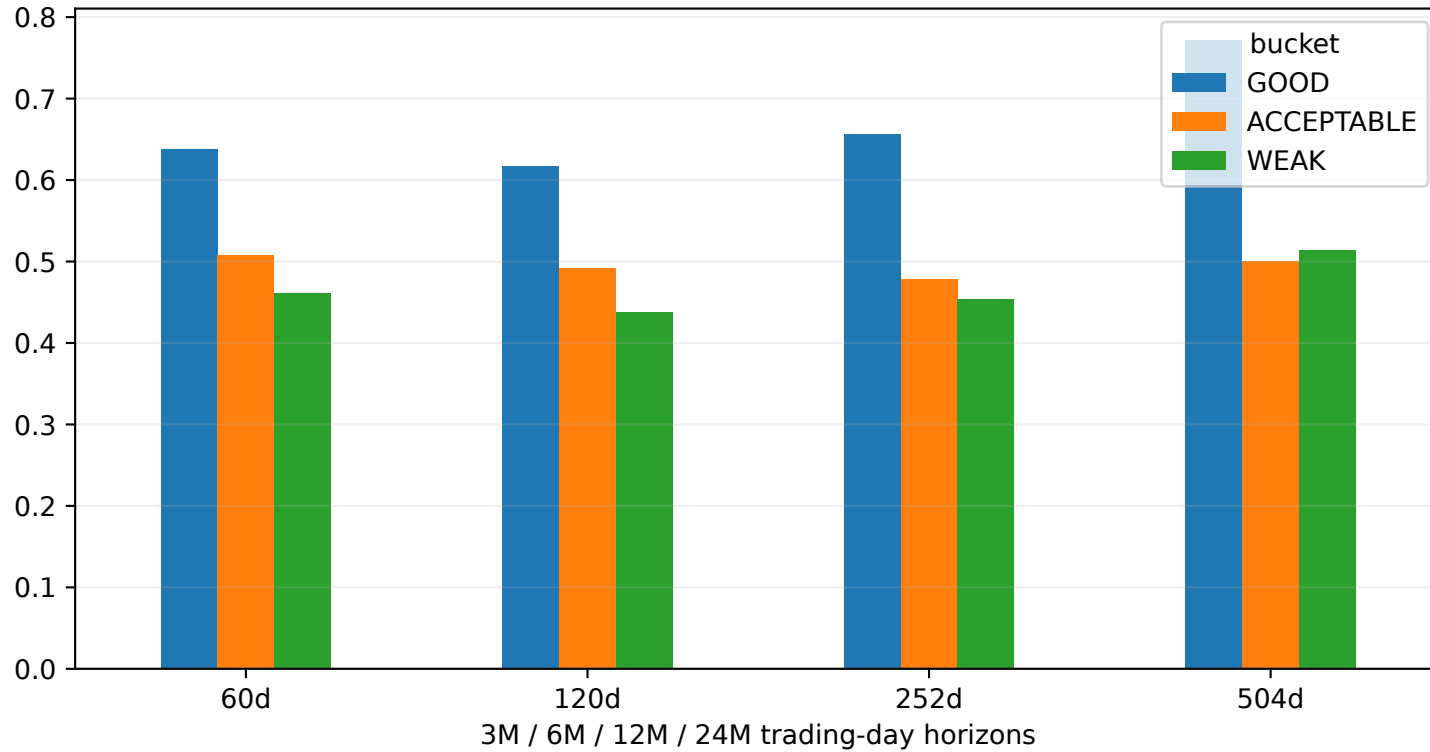
Median stock return



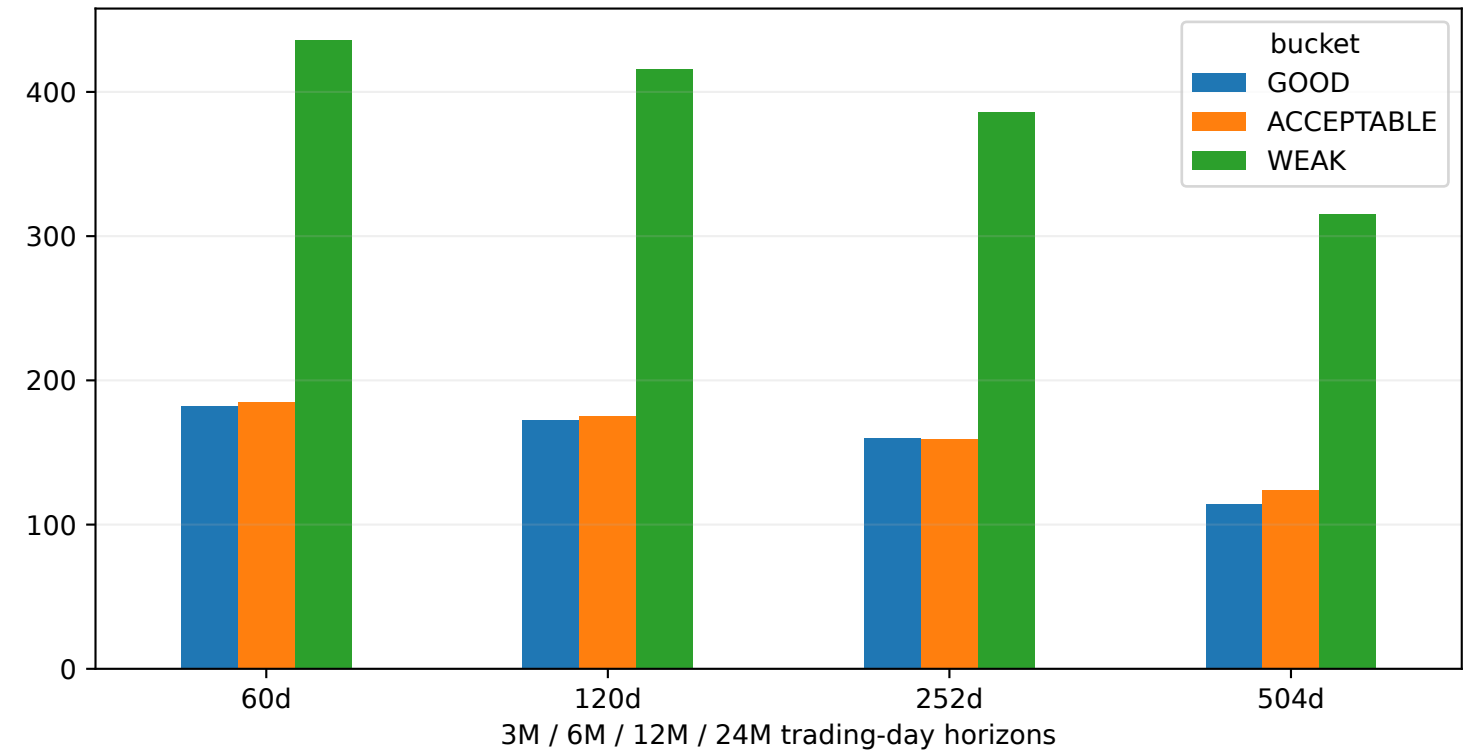
Median excess return vs 0050



Outperform-0050 rate



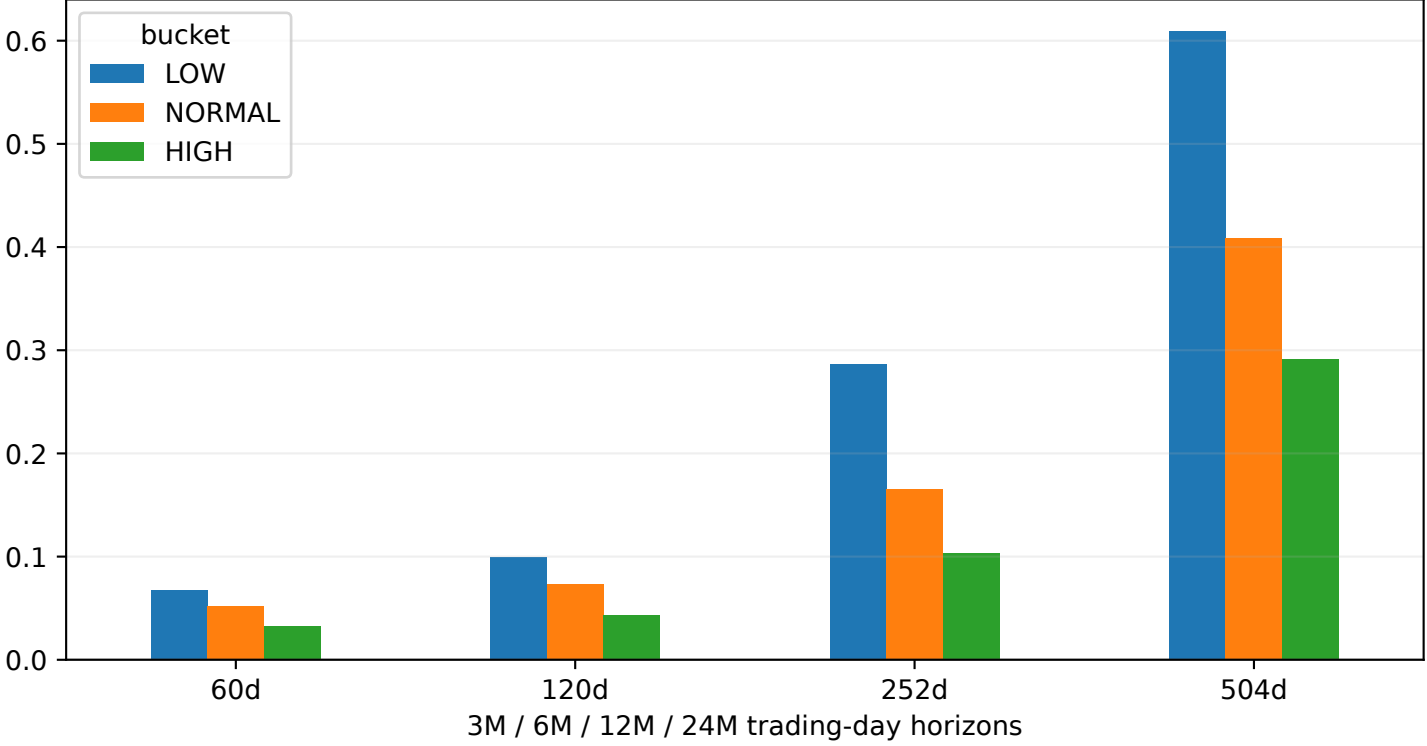
Evaluable observations



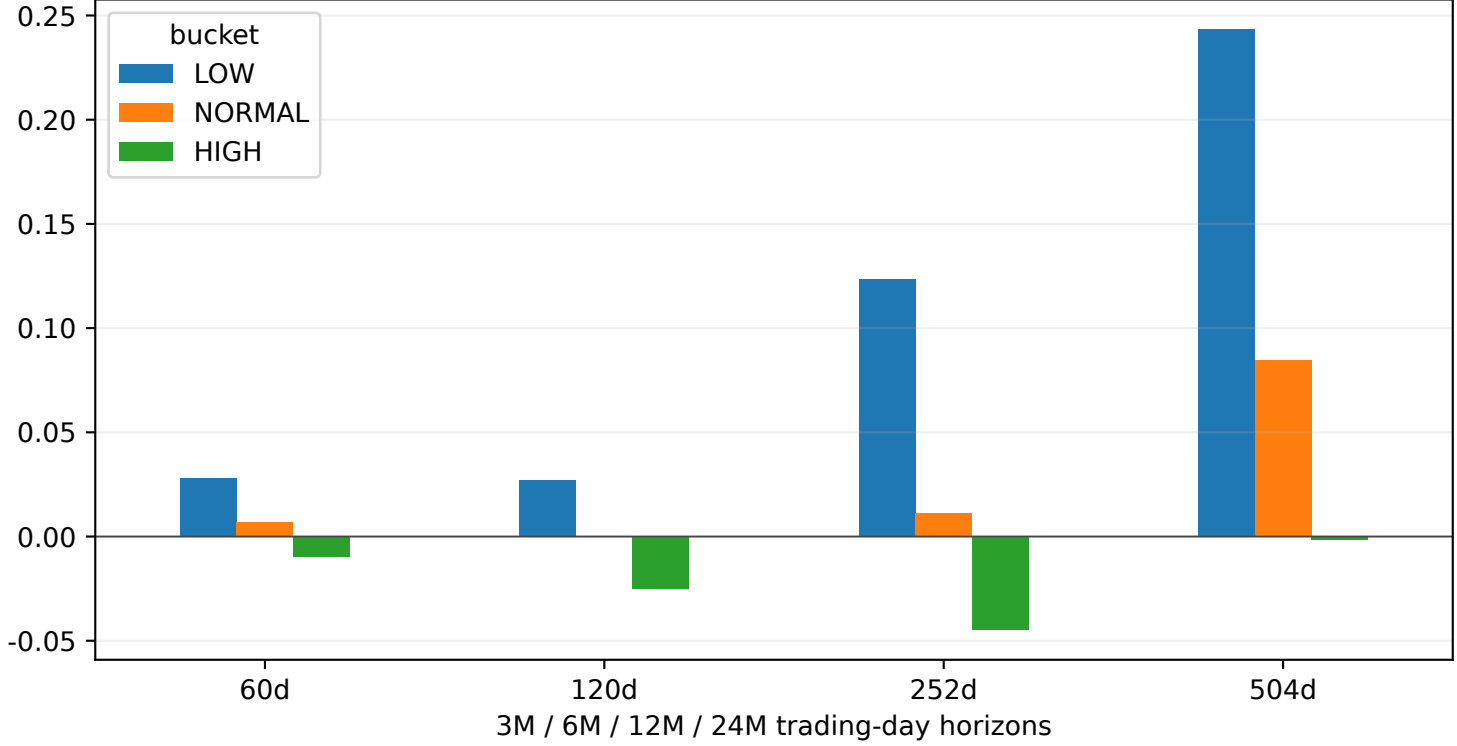
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B4. Valuation Predictive Test

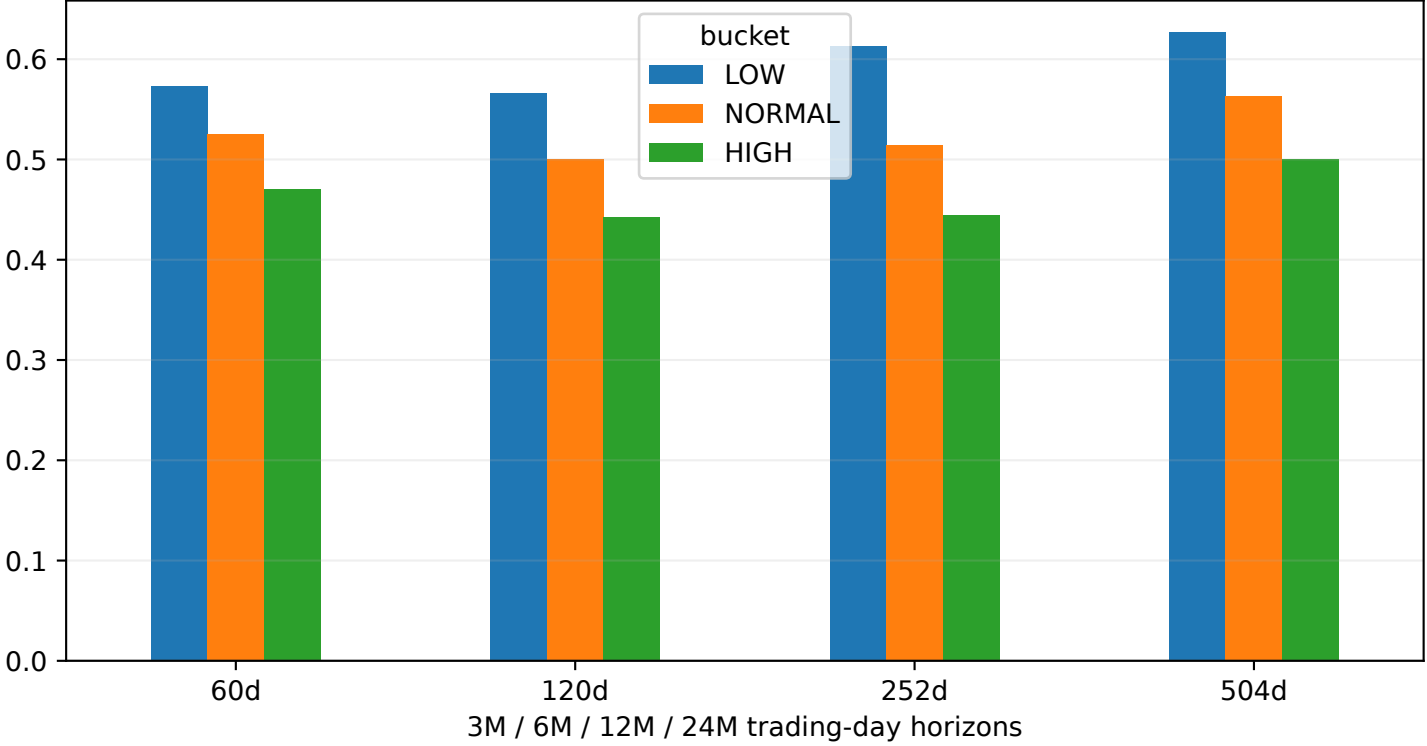
Median stock return



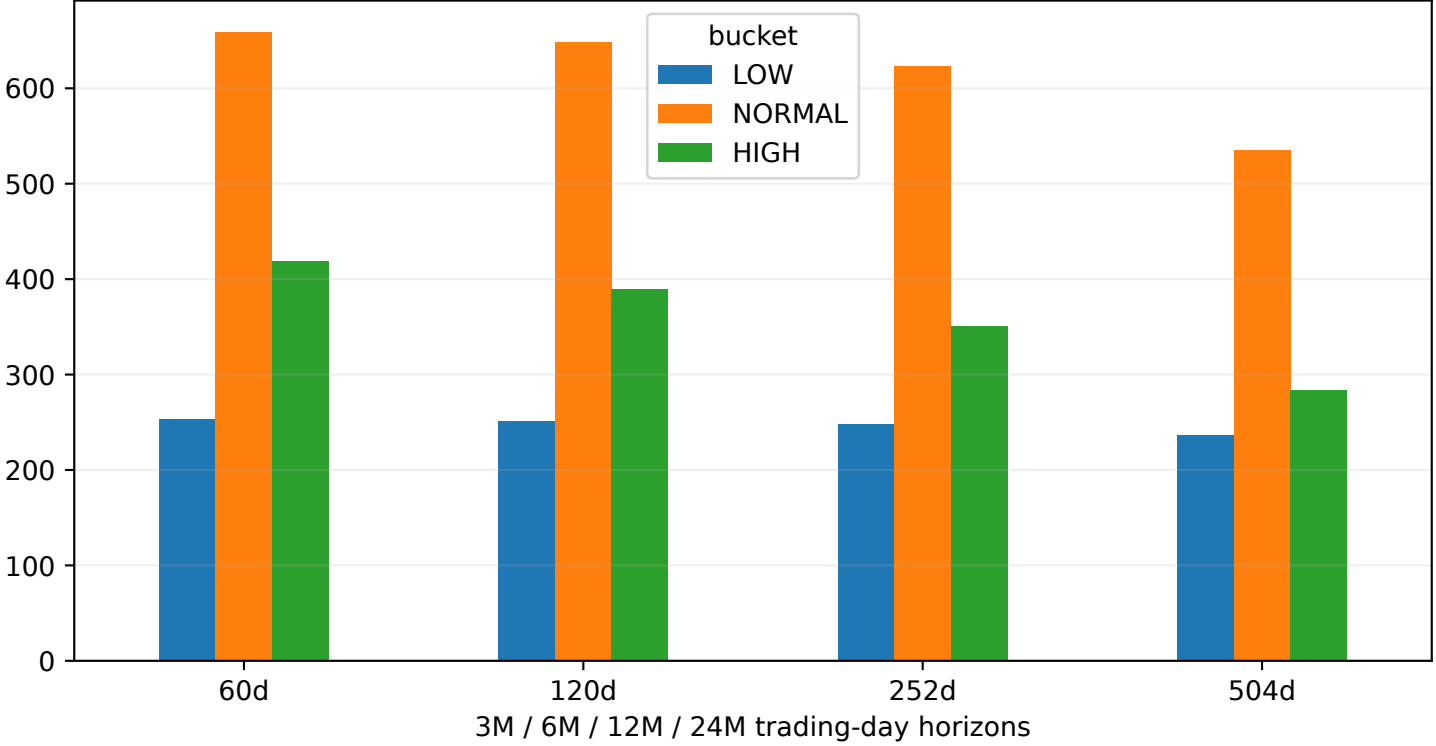
Median excess return vs 0050



Outperform-0050 rate



Evaluable observations



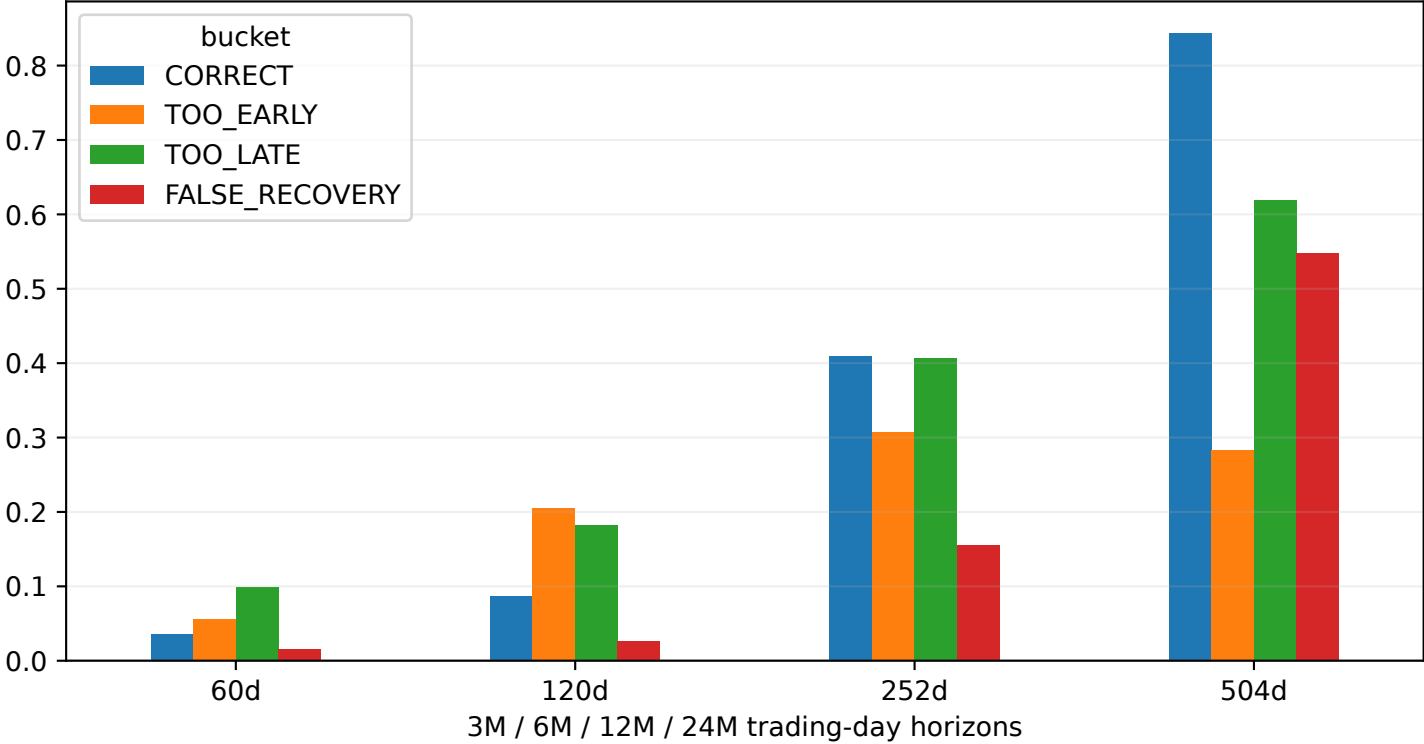
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B5. Supported Categorical Intersections — 12M

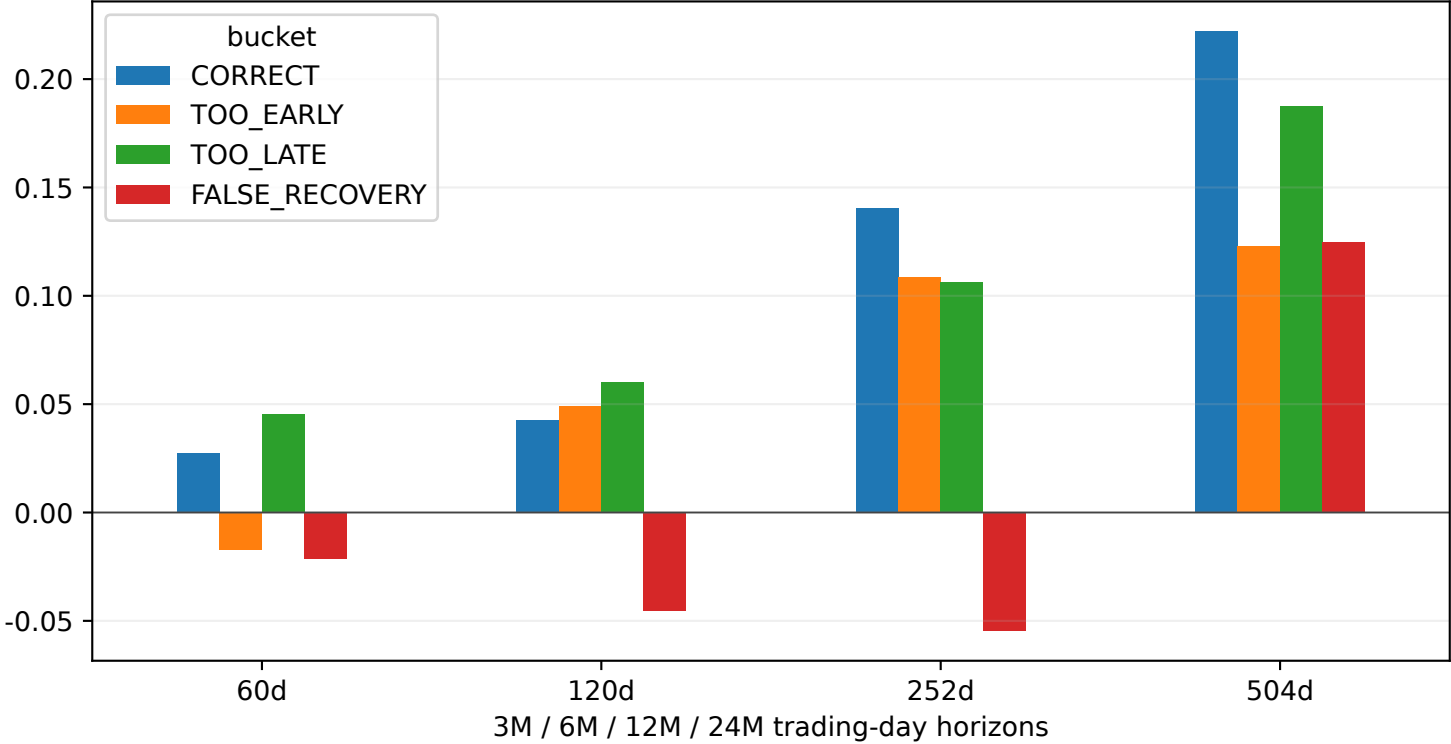
Dimension	Cell	N	Issuers	Median	Median excess	Outperform	Excess lift
QUALITY_X_STATE	GOOD × STABLE	52	15	69.9%	40.1%	71.2%	38.8%
QUALITY_X_STATE	GOOD × IMPROVING	60	14	62.2%	17.3%	80.0%	16.0%
STATE_X_VALUATION	STABLE × LOW	87	24	34.9%	12.4%	62.1%	11.1%
QUALITY_X_VALUATION	ACCEPTABLE × NORMAL	78	23	30.4%	11.3%	57.7%	10.0%
QUALITY_X_VALUATION	WEAK × LOW	79	22	37.3%	10.3%	60.8%	9.0%
STATE_X_VALUATION	IMPROVING × NORMAL	140	33	28.2%	8.6%	62.1%	7.3%
QUALITY_X_VALUATION	GOOD × HIGH	64	14	28.2%	7.8%	64.1%	6.5%
STATE_X_VALUATION	IMPROVING × LOW	61	29	30.8%	7.1%	60.7%	5.8%
STATE_X_VALUATION	IMPROVING × HIGH	102	29	18.6%	5.5%	54.9%	4.2%
QUALITY_X_VALUATION	GOOD × NORMAL	70	19	38.6%	5.4%	60.0%	4.1%
STATE_X_VALUATION	STABLE × NORMAL	143	32	21.7%	2.5%	53.8%	1.2%
QUALITY_X_STATE	ACCEPTABLE × IMPROVING	49	20	22.2%	0.1%	51.0%	-1.2%
QUALITY_X_STATE	WEAK × IMPROVING	109	28	15.4%	-0.3%	49.5%	-1.6%
STATE_X_VALUATION	STABLE × HIGH	48	22	6.3%	-1.2%	47.9%	-2.6%
QUALITY_X_STATE_X_VALUA	WEAK × IMPROVING × NOR	50	22	14.1%	-2.0%	46.0%	-3.3%
QUALITY_X_STATE	ACCEPTABLE × STABLE	40	17	23.5%	-2.7%	47.5%	-4.0%
QUALITY_X_VALUATION	ACCEPTABLE × HIGH	39	11	18.4%	-2.8%	41.0%	-4.1%
QUALITY_X_STATE	WEAK × STABLE	117	25	9.3%	-4.1%	42.7%	-5.4%
QUALITY_X_VALUATION	WEAK × NORMAL	194	27	14.6%	-5.1%	44.8%	-6.4%
QUALITY_X_STATE_X_VALUA	WEAK × STABLE × NORMAL	67	21	11.7%	-9.1%	43.3%	-10.4%
QUALITY_X_STATE_X_VALUA	WEAK × IMPROVING × HIGH	31	13	6.8%	-9.9%	38.7%	-11.2%
QUALITY_X_VALUATION	WEAK × HIGH	94	19	6.7%	-11.4%	36.2%	-12.8%
STATE_X_VALUATION	DETERIORATING × NORMAL	77	29	15.2%	-12.9%	31.2%	-14.2%
QUALITY_X_VALUATION	ACCEPTABLE × LOW	42	10	21.1%	-13.3%	35.7%	-14.6%

B6. TOO_LATE Investment Relevance

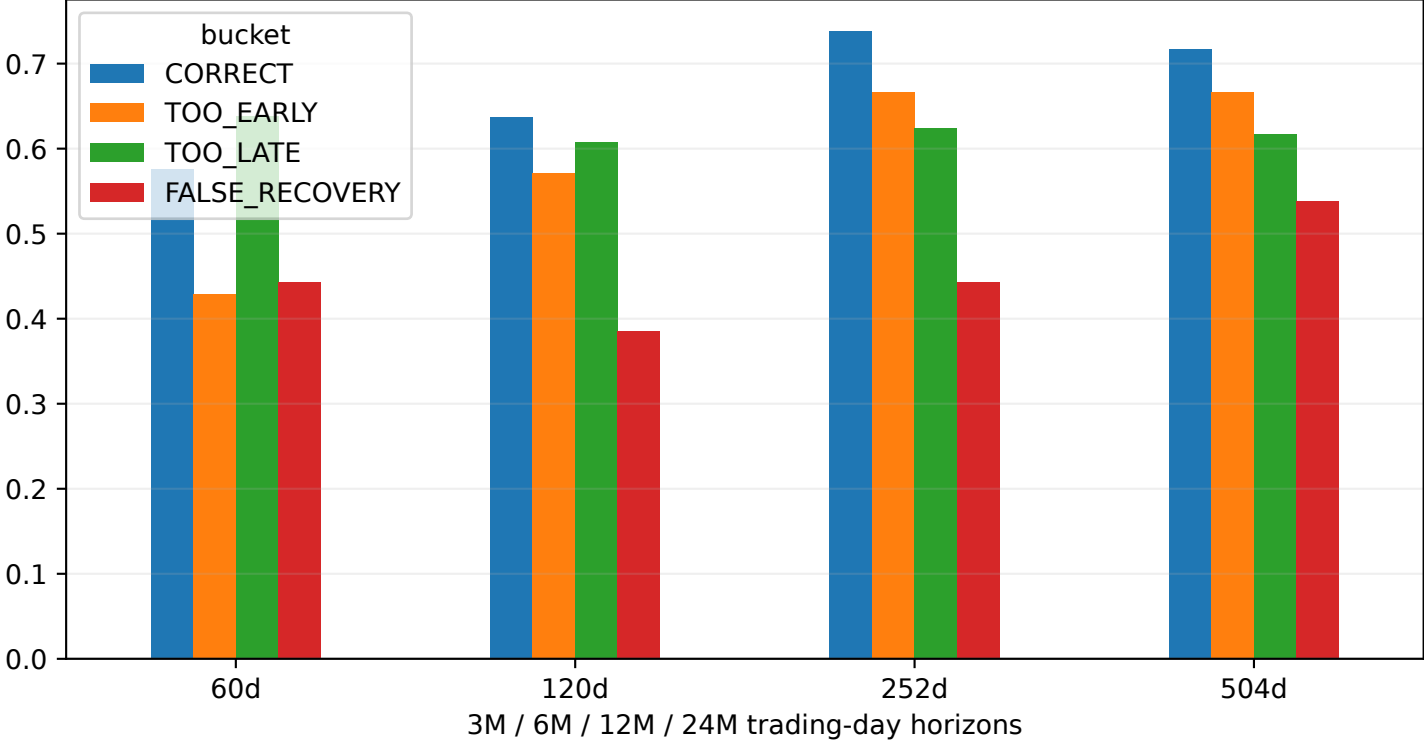
Median stock return



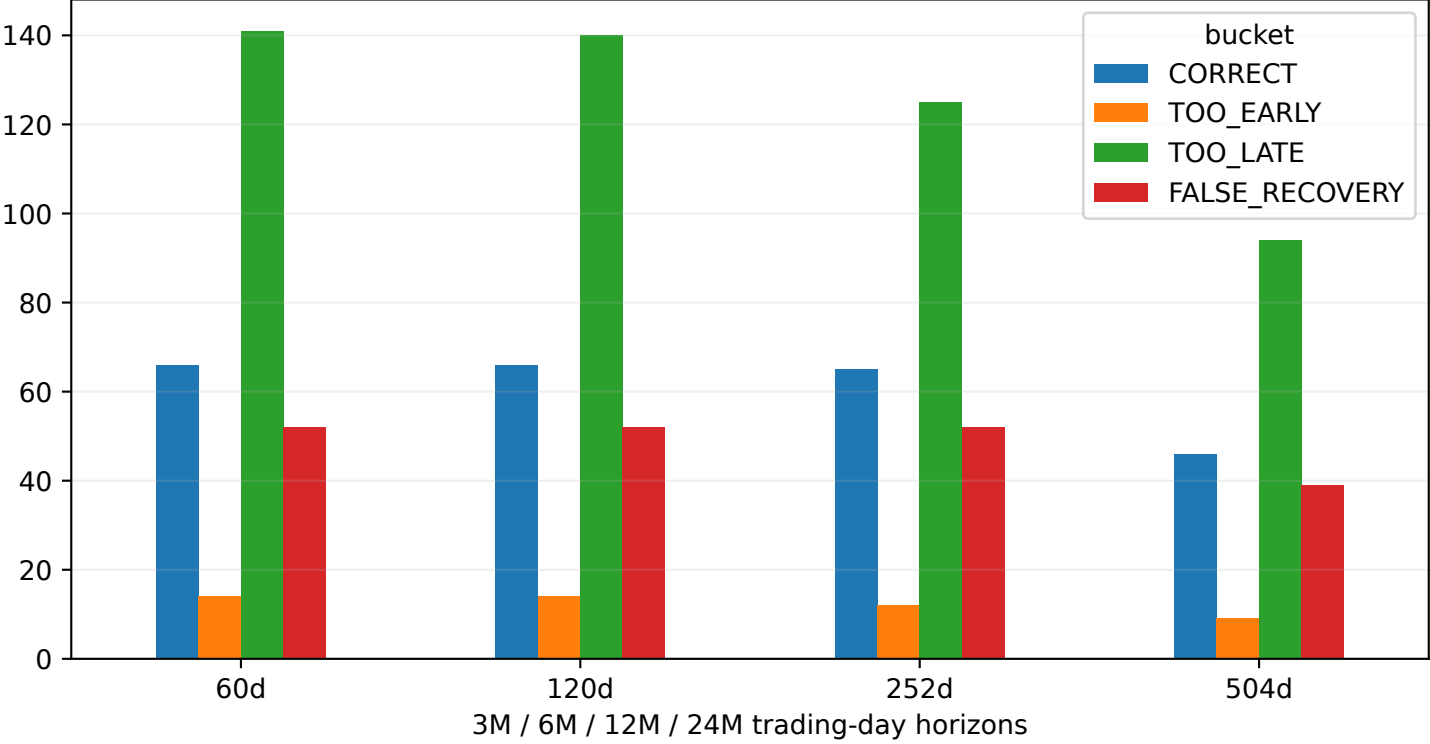
Median excess return vs 0050



Outperform-0050 rate



Evaluable observations



Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B7. Regime Robustness — pre-2023 vs 2023 onward

Regime	State	Horizon	N	Issuers	Median excess	Outperform	Support
PRE_2023	IMPROVING	60d	199	37	1.7%	55.3%	SUPPORTED
PRE_2023	IMPROVING	120d	199	37	2.6%	55.3%	SUPPORTED
PRE_2023	IMPROVING	252d	199	37	7.1%	62.3%	SUPPORTED
PRE_2023	IMPROVING	504d	199	37	17.6%	63.8%	SUPPORTED
PRE_2023	STABLE	60d	180	35	2.5%	58.9%	SUPPORTED
PRE_2023	STABLE	120d	180	35	4.8%	60.0%	SUPPORTED
PRE_2023	STABLE	252d	180	35	5.5%	60.0%	SUPPORTED
PRE_2023	STABLE	504d	180	35	16.5%	61.1%	SUPPORTED
PRE_2023	DETERIORATING	60d	75	31	-2.4%	42.7%	SUPPORTED
PRE_2023	DETERIORATING	120d	75	31	-6.6%	36.0%	SUPPORTED
PRE_2023	DETERIORATING	252d	75	31	-12.1%	32.0%	SUPPORTED
PRE_2023	DETERIORATING	504d	75	31	-12.4%	41.3%	SUPPORTED
2023_ONWARD	IMPROVING	60d	145	36	2.8%	57.9%	SUPPORTED
2023_ONWARD	IMPROVING	120d	134	36	4.2%	56.7%	SUPPORTED
2023_ONWARD	IMPROVING	252d	113	36	6.6%	55.8%	SUPPORTED
2023_ONWARD	IMPROVING	504d	35	23	28.7%	60.0%	SUPPORTED
2023_ONWARD	STABLE	60d	150	32	-2.6%	46.7%	SUPPORTED
2023_ONWARD	STABLE	120d	132	31	-6.3%	41.7%	SUPPORTED
2023_ONWARD	STABLE	252d	110	31	-12.1%	43.6%	SUPPORTED
2023_ONWARD	STABLE	504d	69	25	0.1%	50.7%	SUPPORTED
2023_ONWARD	DETERIORATING	60d	102	35	-1.0%	46.1%	SUPPORTED
2023_ONWARD	DETERIORATING	120d	97	35	-6.8%	42.3%	SUPPORTED
2023_ONWARD	DETERIORATING	252d	86	33	-19.6%	36.0%	SUPPORTED
2023_ONWARD	DETERIORATING	504d	81	32	-18.2%	48.1%	SUPPORTED

B8. Outlier and Concentration Robustness

Scope	Horizon	N	Issuers	Median	Median excess	Outperform	Top-5 contribution
FULL_SAMPLE	60d	344	37	6.7%	2.3%	56.4%	5.7%
EX_TSMC	60d	330	36	6.8%	2.5%	56.1%	5.7%
TOP_1_RETURN_REMOVED	60d	344	37	6.7%	2.3%	56.4%	5.7%
TOP_5_RETURNS_REMOVED	60d	343	37	6.7%	2.2%	56.3%	5.7%
FULL_SAMPLE	120d	333	37	12.3%	3.0%	55.9%	6.4%
EX_TSMC	120d	319	36	12.4%	3.8%	55.5%	6.4%
TOP_1_RETURN_REMOVED	120d	333	37	12.3%	3.0%	55.9%	6.4%
TOP_5_RETURNS_REMOVED	120d	333	37	12.3%	3.0%	55.9%	6.4%
FULL_SAMPLE	252d	312	37	27.3%	7.0%	59.9%	6.1%
EX_TSMC	252d	298	36	26.9%	7.1%	58.7%	6.1%
TOP_1_RETURN_REMOVED	252d	312	37	27.3%	7.0%	59.9%	6.1%
TOP_5_RETURNS_REMOVED	252d	312	37	27.3%	7.0%	59.9%	6.1%
FULL_SAMPLE	504d	234	37	63.8%	19.9%	63.2%	5.2%
EX_TSMC	504d	224	36	63.0%	17.9%	62.9%	5.2%
TOP_1_RETURN_REMOVED	504d	234	37	63.8%	19.9%	63.2%	5.2%
TOP_5_RETURNS_REMOVED	504d	231	37	63.0%	18.1%	62.8%	5.2%

B9. Overlapping-Window Audit

Horizon	Eligible	Overlapping obs	Ratio	Pairs	Same issuer	Cross issuer/calendar
60d	1502	1502	100.0%	48932	441	48491
120d	1457	1457	100.0%	101253	2042	99211
252d	1390	1390	100.0%	210620	5148	205472
504d	1219	1219	100.0%	330304	8386	321918

B10. Evidence Grade and Limitations

MOPS PIT coverage

Frozen Stage A financial timeline: 2,067 observations; 2,053 MOPS exact (99.3%) and 14 explicit available-date proxies. Stage B performs zero MOPS network requests and does not remap any filing date.

Evidence grade

STRONG — EXPLORATORY_PREDICTIVE_ASSOCIATION. The deterministic rubric was registered in code and documentation before this run.

Uncertainty

Every summary retains issuer-clustered and entry-quarter-clustered mean uncertainty plus deterministic cluster-bootstrap median intervals. Ordinary IID confidence is not used as the evidentiary basis.

Overlap

The overlap artifact counts observations and pairs whose forward windows intersect. The 1,563 signal records are not treated as 1,563 independent trials.

Limitations

CURRENT_CONSTITUENTS_ONLY; results are not survivorship-bias-free historical 0050 performance. Repeated issuer observations and overlapping windows are not IID. Issuer- and entry-quarter-clustered uncertainty does not eliminate all common-regime dependence. Adjusted-close paths are close-to-close; excursions are not intraday MFE/MAE. Categorical intersections are exploratory and are not scores or tuned trading rules. Financial issuers are excluded from all predictive evidence.

Prohibited claims

No buy/sell recommendation, target price, expected-return promise, historical-0050 strategy performance, causal claim, composite score, or parameter optimization.